

QUESTION BANK CAE-2

Subject:[Operating Systems]

Class: SYAI[A] (UCAIL202)

Blooms Taxonomy level

1: Remember, 2: Understand, 3: Apply, 4: Analyse, 5: Evaluate, 6: Create

Q. No.	Question	Marks	BL	CO	PO
UNIT III					
1	What is the critical section? Give semaphore solution for producer-consumer problem	4	1	2	1
2	Explain Semaphore with its type.	4	3	2	1
3	What is busy waiting with respect to process synchronization?	4	1	2	1
4	Explain how semaphore solves problem of synchronization	3	2	2	1
5	Explain following terms: i) Critical Section. ii) Race Condition.	3	2	2	1
6	Describe in brief different IPC mechanisms	3	2	2	1
7	Describe the requirements of the synchronizing mechanism.	4	2	2	2
8	Discuss the Critical Section Problem with pseudo code.	4	2	2	2
9	Explain Co-operating processes in an Inter- process communication (IPC) mechanism.	4	2	2	1

10	Illustrate Peterson’s software-based solution to the critical section problem.	4	3	2	1																																																																																											
UNIT IV																																																																																																
11	List four Deadlock prevention conditions and explain the following terms. 1) Removal of “No preemption” condition.	4	1	3	1																																																																																											
12	Describe necessary conditions for deadlock.	3	2	3	2																																																																																											
13	Explain with an appropriate example, how resource allocation graph determines a deadlock.	4	2	3	2																																																																																											
14	Explain deadlock prevention in detail.	4	2	3	1																																																																																											
15	Explain deadlock avoidance using banker’s algorithm in detail.	4	2	3	2																																																																																											
16	Consider the following snapshot <table border="1"><thead><tr><th></th><th colspan="4">ALLOCATED</th><th colspan="4">MAX</th><th colspan="4">AVAILABLE</th></tr><tr><th></th><th>A</th><th>B</th><th>C</th><th>D</th><th>A</th><th>B</th><th>C</th><th>D</th><th>A</th><th>B</th><th>C</th><th>D</th></tr></thead><tbody><tr><td>P0</td><td>0</td><td>0</td><td>1</td><td>2</td><td>0</td><td>0</td><td>1</td><td>2</td><td>1</td><td>5</td><td>2</td><td>0</td></tr><tr><td>P1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>7</td><td>5</td><td>0</td><td></td><td></td><td></td><td></td></tr><tr><td>P2</td><td>1</td><td>3</td><td>5</td><td>4</td><td>2</td><td>3</td><td>5</td><td>6</td><td></td><td></td><td></td><td></td></tr><tr><td>P3</td><td>0</td><td>6</td><td>3</td><td>2</td><td>0</td><td>6</td><td>5</td><td>2</td><td></td><td></td><td></td><td></td></tr><tr><td>P4</td><td>0</td><td>0</td><td>1</td><td>4</td><td>0</td><td>6</td><td>5</td><td>6</td><td></td><td></td><td></td><td></td></tr></tbody></table> Answer the following questions using banker’s algorithm: a) What are the contents of the matrix end? b) Is the system in safe state? c) If request for process p1 arrives for (0,4,2,0) .Can the request be granted immediately?		ALLOCATED				MAX				AVAILABLE					A	B	C	D	A	B	C	D	A	B	C	D	P0	0	0	1	2	0	0	1	2	1	5	2	0	P1	1	0	0	0	1	7	5	0					P2	1	3	5	4	2	3	5	6					P3	0	6	3	2	0	6	5	2					P4	0	0	1	4	0	6	5	6					4	3	3	2
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P4	0	0	1	4	0	6	5	6																																																																																								

17	Consider a system with a total of 150 minutes of memory allocated to three processes as shown. Apply banker's algorithm to determine whether it would be safe to grant each of the following requests. If yes-Indicate sequence of termination that could be possible. If no- Show reduction of resulting allocation table. 1. A 4th process is arrived at with a maximum need of 60 and initial need of 25 units. 2. A 4th process is arrived at with a maximum need of 60 and initial need of 35 units. <table><tr><td>Process</td><td>Max</td><td>Hold</td></tr><tr><td>P1</td><td>70</td><td>45</td></tr><tr><td>P2</td><td>60</td><td>10</td></tr><tr><td>P3</td><td>60</td><td>15</td></tr></table>	Process	Max	Hold	P1	70	45	P2	60	10	P3	60	15	4	3	3	2
Process	Max	Hold															
P1	70	45															
P2	60	10															
P3	60	15															
18	Apply the deadlock free solution for a Dining Philosopher problem	4	3	3	2												
19	Write a short note on following i) Resource allocation graph ii) Mutual exclusion	4	2	3	2												
20	How does a resource allocation graph help to deadlock? Write the necessary conditions of the deadlock to be occurred.	4	2	3	2												

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